

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (EC) Examination

November 2018

EC342 VLSI Technology and Design

Date: 22/11/2018, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the questions below. [07]

(a) What is polarity of threshold voltage of NMOS transistor? [01]

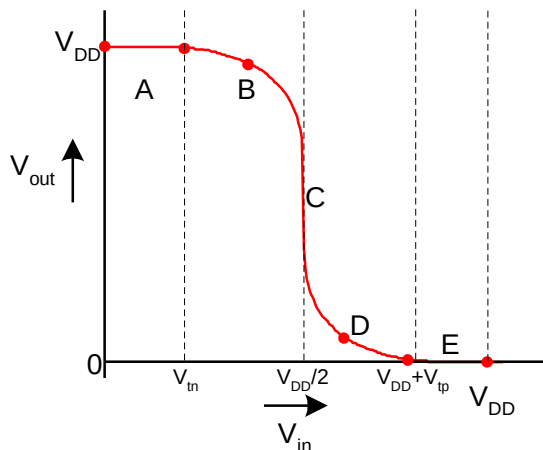
- (i) Positive (ii) Negative (iii) Either Positive or Negative

(b) “Pinch-off” because of channel length modulation occurs in which region of operation? [01]

- Subthreshold
- Saturation
- Linear
- Both (b) and (c)

(c) Because of short channel effect, the _____ of a MOSFET _____ (increases/decreases) with increasing values of L . [02]

(d) For each of the regions shown below in the DC transfer curve, identify the regions of operation for the PMOS and NMOS. Complete the table [03]



Region	nMOS	pMOS
A		
B		
C		
D		
E		

Q – 2. Answer any two questions. [14]

(a) Consider an NMOS transistor with $W/L = 4/2$. In this process, the gate oxide thickness is 100 Å and the mobility of electrons is $350 \text{ cm}^2/\text{V.s}$. The threshold voltage is 0.7 V. Plot I_{ds} vs V_{ds} for $V_{gs} = 0, 1, 2, 3$. [07]

- (b) Calculate the threshold voltage V_{T0} under zero bias for a polysilicon gate MOS transistor [07]
with following parameters: Substrate doping density $N_A=10^{16} \text{ cm}^{-3}$, polysilicon gate
doping density $N_D = 10^{20} \text{ cm}^{-3}$, $t_{ox} = 500 \text{ \AA}$, Oxide interface charge density
 $N_{ox} = 2 \times 10^{10} \text{ cm}^{-3}$. Take $\epsilon_0=8.85 \times 10^{-14} \text{ F/Cm}$, $\epsilon_{si}=11.7 \times \epsilon_0 \text{ F/Cm}$, $\epsilon_{ox} = 3.97 \times \epsilon_0$
 F/Cm .

- (c) Calculate NM_L (Noise margin, low) of a CMOS inverter having following parameters: [07]
 $V_{DD}=3.3\text{V}$, $V_{to,n}=0.6\text{V}$, $V_{to,p}= -0.7\text{V}$, $K_n = 400 \mu\text{A/V}^2$, $K_p = 160 \mu\text{A/V}^2$.

Q – 3. Answer any two questions. [14]

- (a) i. Synchronous Clocking in sequential circuits leads to several problems. Justify the [04]
statement with the help of precharge evaluate logic.
ii. Tabulate different parameters (Current, Power, Power density) for Constant voltage [03]
Scaling and Full Scaling.
- (b) i. Discuss oxide capacitance in detail. [04]
ii. Explain the following terms using necessary waveforms and circuit diagram for a [03]
CMOS inverter:
- Rise Time, Fall Time, τ_{PHL} , τ_{PLH}
- (c) Threshold voltage drops at the output in an inverter can be overcome using _____ [07]
- i. Voltage Bootstrapping circuit
 - ii. Built-in self test
 - iii. Voltage Doubler
 - iv. Voltage tripler
- Justify with the help of necessary equations and circuit diagram.

SECTION – II

Q - 4 Answer the following questions. [07]

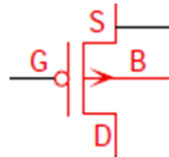
- (a) How threshold voltage of a fabricated MOSFET can be varied? [01]
- (b) V_{OL} of Depletion load NOR gate _____ (increases/decreases) on increasing FAN -in [01]
of the gate.
- (c) What is the minimum number transistor(s) required to implement following design (F) [01]
using CMOS Logic? $F = A + BA + BAC + BCAD + ABCDE + FEDCBA$
- i. 21
 - ii. 42
 - iii. 18
 - iv. None of these
- (d) Which of the following statement(s) is/are false for S-R (Set Reset) Latch? [01]
- P:** Active High S-R Latch is NAND based Latch.
- Q:** Race condition occurs for $S=0$ & $R=0$ in NOR Based S-R Latch.

- i. Only Statement P
- ii. Only Statement Q
- iii. Both Statement P & Q
- iv. Neither Statement P nor Q

(e) CMOS inverter is designed using NMOS and PMOS transistor having threshold voltage 0.7 and -0.7 respectively. What is the minimum supply voltage that can be used in this design without affecting the operation of inverter? [01]

- i. 0.7 V
- ii. -0.7 V
- iii. 1.4 V
- iv. Can't be determined

(f) [01]



If drain and source potential are 4V and -5V respectively. What should be the maximum gate potential so that the PMOS device ($V_{T0} = -0.7V$) remains ON.

- i. 3.3V
- ii. 4.7V
- iii. -4.3V
- iv. -5.7V

(g) What do you mean by Locality in VLSI Design? [01]

Q - 5 Answer the questions below (Any Two). [14]

(a) Explain the process of NMOS fabrication with suitable diagram. [07]

(b) Implement the following: [07]

- i. 4:1 MUX using CMOS Transmission gate
- ii. Two input (one bit) Inequality Detector circuit using pseudo-NMOS Logic

(c) Design D Flip – Flop (Positive Edge triggered) using CMOS transmission gates and CMOS Inverter and answer the following. [07]

- i. Number of clock signals
- ii. Number of transmission gates and CMOS Inverter
- iii. Explain the operation

Q - 6 Answer the questions below (Any Two). [14]

(a) A CMOS NAND gate having fan-in of 2 is to be designed. If all the NMOS transistors have equal transconductance parameter (k_N) and threshold voltage (V_{TN}). All PMOS transistors have equal transconductance parameter and threshold voltage k_N & V_{TP} respectively. Which of the following Design Engineer will give threshold voltage of CMOS NAND gate as $V_{DD}/2$? Justify your Answer. [07]

- Design Engineer **Shaurya** : $V_{TP} = V_{TN}$
- Design Engineer **Hausla** : $k_P = 2k_N$
- Design Engineer **Saahas** : $V_{TP} = V_{TN}$ & $k_N = 4k_P$

- Design Engineer **Jazba** : $V_{TP} = V_{TN}$ & $k_p = 4k_n$

Use this NAND gate to implement CMOS clocked S-R Latch.

- (b) What is the need of Euler Path in stick diagram? [07]

Realize the CMOS logic circuit for $Z = \overline{(D + E + A)(B + C)}$. Find the equivalent

CMOS inverter circuit for simultaneous switching of all inputs, assuming that $(W/L)_p = 15$ for all pMOS transistors and $(W/L)_n = 10$ for all nMOS transistors

- (c) Explain MOS capacitor under external bias with energy band diagrams (consider P-type substrate). [07]